

## TB-500 Available for Research Use Only

(2-2.5 mg 2×/wk × 4–6 wk, then 2-2.5 mg 1×/wk until healed)

TB-500 (thymosin-β4 derivative) is widely used for tissue repair, wound healing, and neurological healing with proven success in veterinary use and widespread positive reports among human practitioners. Its short-term safety profile is good, but its long-term use carries a significant caution due to mechanistic cancer risks, limiting its use to acute repair only, and with an absolute contraindication for subjects with elevated cancer risk.

  Probably safe (some human evidence)

  Probably effective (strong animal signals and/or some positive human signals)

**C** Angiogenic – caution for subjects at elevated risk for cancers, Do NOT use with active cancer

*Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.*

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**TB-500** is the synthetic fragment of **Thymosin Beta-4 (Tβ4)** designed to **replicate an active region of the native peptide**, retaining its key biological functions including **actin sequestration, promotion of cell migration, angiogenesis, and anti-inflammatory signaling**.

The evidence for TB-500's long term human safety for **non-continuous use** is just moderate because safety **support comes from mechanistic, in-vitro, and animal tolerability data plus anecdotal clinical use**, but no peer-reviewed subcutaneous human safety trials or long-term surveillance are available. **Evidence for long-term safety for continuous use is weaker due to mechanistic pro-tumor concerns.**

The evidence for TB-500's efficacy for improved wound healing is high because **convergent mechanistic pathways and reproducible animal wound-healing studies provide a strong signal of benefit**, though controlled human efficacy trials for subQ TB-500 are lacking.

### Standard Biphasic Dosing & Patterns

We have no evidence-based dose for TB-500. Instead, we rely on a mechanistic understanding and community protocols for subcutaneous TB-500 for acute injury repair, which almost universally follow a biphasic structure: an initial loading phase to achieve tissue saturation, followed by a maintenance phase. This approach is consistent with the peptide's mechanism, as Tβ4-mediated G-actin sequestration and cell migration benefit from an initial period of supraphysiological concentrations. Three dosing strategies emerge for different purposes:

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- **Mild injury & general recovery** (conservative), Loading: 2 mg 2×/wk × 4 wk, maintenance: 2 mg 1×/wk during continued recovery
- **Musculoskeletal injury** (Standard), Loading: 2.5 mg 2×/wk × 4–6 wk, maintenance: 2–2.5 mg 1×/wk during continued recovery
- **Severe / cardiac** (high load), Loading: 3 mg 2×/wk × 6 wk, maintenance: 2.5 mg 1×/wk during continued recovery, clinical oversight advised

Given TB-500s half-life, twice/week dosing makes sense, however more frequent dosing is also effective as long as per week dosing matches, e.g. 2 mg 2×/wk or 600 mcg /daily.

All these TB-500 protocols are intended to **complete at the end of injury recovery**. However, **continuous use is discussed to support scar and fibrosis remodeling, continuous training recovery, general wellness, longevity, hair follicle activation**. A common dose for **continuous use might be 2mg/week**, although lower and much higher doses are discussed. We have limited safety data for TB-500, but the data we have suggests even at supra-physiological doses there are **no toxicities or serious adverse events**. However, we have only short-term data, and there are **legitimate mechanistic concerns about TB-500's acceleration of angiogenesis supporting tumor growth as a long-term risk**. Any use of continuous TB-500 beyond acute injury repair should **take into account long-term cancer risks** which vary by subject.

### Common Stacking

TB-500 is frequently **combined with BPC-157** (Body Protection Compound-157) in community protocols, as the two peptides are thought to have complementary mechanisms: TB-500 provides systemic tissue repair via actin regulation and angiogenesis, while BPC-157 acts more locally on gut-healing pathways and nitric oxide modulation. No clinical trials have evaluated this combination.

### Benefits

Almost all the data we have comes from research on **Thymosin Beta-4**. It is a mechanistic interpretation to use that data to describe benefits for TB-500.

- **Tissue Repair and Wound Healing** - accelerates wound closure through upregulation of actin, promoting cell migration across damaged tissue matrices.
- **Cardiovascular Protection** - promotes cardiac cell survival and reduces scar size after myocardial infarction
- **Anti-Inflammatory Effects** - downregulates pro-inflammatory cytokines, modulates NF-κB signaling, and inhibits interferon-γ production in T-cell/monocyte co-cultures. Evans et al. (2013) showed that Tβ4-sulfoxide triggers depletion of inflammatory macrophages at injury sites, promoting resolution of inflammation and reduced scarring.

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- **Musculoskeletal Recovery** - TB-500 has been widely used in equine veterinary medicine for tendon and ligament repair, providing real-world evidence of efficacy in soft-tissue musculoskeletal injury. Its systemic distribution after subcutaneous injection allows it to reach injury sites throughout the body without requiring local administration.
- **Neurological Recovery** - demonstrated improved functional recovery after traumatic brain injury through promotion of neural stem cell proliferation and reduced neuroinflammation. Promoted remyelination in multiple sclerosis models by supporting oligodendrocyte differentiation.
- **Hair Follicle Activation** - promotes hair growth in through stem cell activation in hair follicles, though human clinical data for this indication remains limited.

### Indications

- Soft-tissue injury recovery (tendon, ligament, muscle strains and tears)
- Post-surgical wound healing acceleration
- Chronic tendinopathy and overuse injuries
- Corneal wounds, dry eye disease, and neurotrophic keratitis (RGN-259 formulation)
- Cardiac protection and regeneration post-myocardial infarction
- Dermal wound healing including chronic ulcers (venous stasis ulcers)
- Neurological injury recovery (traumatic brain injury, demyelinating conditions)
- Hepatic and renal fibrosis reduction
- General anti-inflammatory support in musculoskeletal contexts
- Alopecia and hair follicle stimulation (limited evidence)

### Contraindications

- **Active malignancy or history of cancer:** Tβ4 promotes angiogenesis and cell migration, which could theoretically accelerate tumor growth or metastasis. This is the primary and most significant theoretical concern.
- **Pregnancy and breastfeeding:** No reproductive safety data is available in humans. Use is contraindicated due to unknown teratogenic potential.
- **Known hypersensitivity:** Allergy to Thymosin Beta-4 or any formulation excipients (e.g., mannitol in lyophilized preparations).
- **Pediatric populations:** No pediatric safety data exists. Use in children and adolescents is contraindicated.
- **Active systemic infections requiring treatment:** Theoretical concern regarding immune modulation during active infection. Tβ4 has immunomodulatory properties that may interfere with the immune response to pathogens.

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### Side Effects

- Temporary fatigue (self-resolving; dose at bedtime)
- Headache
- Mild nausea (rare)
- Flu-like symptoms due to immune activation (rare, self-resolving)

### Drivers of Diminished Response and Mitigation Strategies

For **Receptor desensitization** the risk is low because TB-500 is a thymosin beta 4 fragment that modulates actin dynamics rather than a classical receptor agonist in human studies. For **Neutralizing antibodies** the risk is moderate because synthetic peptide fragments can provoke immune responses and published human immunogenicity data are limited. For **downstream pathway adaptation** repeated stimulation of migration and angiogenesis pathways could lead to attenuated intracellular signaling over time. For **System Level Physiological Adaptation** tissue remodeling homeostasis may reduce marginal benefit with chronic exposure. **Overall likelihood of waning is moderate.**

Mitigation is **short cycles**.

### Biological Mechanism

We extrapolate the biology of TB-500 from Thymosin Beta-4.

The primary molecular function of Tβ4 is **high-affinity binding to globular actin (G-actin)**, thereby regulating the **polymerization of actin filaments (F-actin)**. This sequestration of G-actin monomers controls cytoskeletal dynamics critical for **cell shape, division, motility, and migration**. By maintaining a dynamic pool of available actin monomers, Tβ4 enables rapid cytoskeletal remodeling when cells must respond to injury signals.

The biological activity of Tβ4 is determined by distinct **functional domains within its amino acid sequence**. The **first four amino acids regulate anti-inflammatory and antifibrotic effects**. **Amino acids 1–15 inhibit apoptosis** and reduce toxicity-induced cellular damage. The **active fragment encoded by amino acids 17–23 (the Ac-SDKP tetrapeptide region)** triggers **angiogenesis and hair follicle growth**. This tetrapeptide sequence is the core bioactive motif replicated in synthetic TB-500.

Tβ4 promotes **angiogenesis** by enhancing **endothelial progenitor cell (EPC) viability** and triggering the **proliferation and migration of endothelial cells** as well as **formation of capillary-like structures**. This neovascularization improves oxygen and nutrient delivery to damaged tissues, accelerating repair.

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At the signaling level, Tβ4 interacts with **PINCH-1**, a key scaffolding protein in integrin-linked **kinase (ILK) signaling**, connecting Tβ4 activity to broader **cell adhesion and survival pathways**. Tβ4 modulates **NF-κB inflammatory signaling**, **downregulates pro-inflammatory cytokines**, and in cardiac tissue, has been shown to **regulate ROCK1 expression via miR-139-5p modulation** (Maar et al., 2025), contributing to **reduced pathological scarring** following myocardial infarction.

In cardiac models specifically, Tβ4 has been shown to **activate epicardial progenitor cells**, cells capable of differentiating into **new cardiomyocytes**. The sulfoxide form of Tβ4 (Tβ4-SO) attenuates **inflammatory macrophage infiltration** at injury sites by inhibiting **interferon-γ production** and stimulating **superoxide-mediated monocyte clearance**.

TB-500's **low molecular weight (~4,963 Da)** enables **systemic distribution after subcutaneous injection**, allowing a single injection to exert effects at distant injury sites throughout the body. This systemic bioavailability distinguishes TB-500 from peptides requiring local administration and underpins its utility for diffuse or multi-site tissue repair.

### TB-500 and Cancer Risk

**TB-500 has no evidence of causing cancer in healthy tissue, but its biological actions overlap with several pathways that tumors can exploit, which is why the cancer-risk discussion exists.** In animal and in-vitro studies, full-length Thymosin β4 is **upregulated in many cancers** and is involved in **cell migration, angiogenesis, epithelial–mesenchymal transition (EMT), and cytoskeletal remodeling**, all processes that malignant cells hijack to become more invasive. TB-500 preserves the **actin-binding and pro-angiogenic domains** of Thymosin β4, meaning it retains the functions most relevant to **cell motility and new blood-vessel formation**. While **no study has shown TB-500 initiates cancer**, and no in-vivo model has demonstrated spontaneous tumor formation or accelerated tumor growth from TB-500 exposure, we also have **no long-term safety data showing TB-500 is not associated with increased cancer risk**.

There is other evidence that might reduce this mechanistic pro-tumor message: In a myocardial infarction model, **Bock-Marquette et al.** showed that thymosin β4 promoted cardiomyocyte survival and tissue repair **without inducing dysplasia, abnormal proliferation, or tumor formation**, demonstrating neutral oncologic behavior in a high-proliferation context. In corneal wound-healing studies, **Sosne et al. (2002)** found that thymosin β4 accelerated epithelial migration and reduced inflammation **with no neoplastic changes or abnormal proliferative lesions** in treated ocular tissue. In a follow-up corneal regeneration study, **Sosne et al. (2004)** reported enhanced epithelial repair driven by thymosin β4 **without evidence of hyperproliferation or tumor-like growth**, confirming neutral behavior in a second ocular model. In dermal wound-healing experiments, **Philp et al.** demonstrated that thymosin β4 improved

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re-epithelialization and reduced inflammatory cytokines **while maintaining normal tissue architecture and showing no tumorigenic activity** in skin. In a cardiac progenitor mobilization study, **Smart et al.** showed that thymosin  $\beta$ 4 activated epicardial progenitor cells and supported cardiomyocyte regeneration **without uncontrolled proliferation or neoplastic transformation**, indicating regenerative but non-oncogenic signaling.

The pro-tumor concern is **theoretical**, not demonstrated: if an individual already has an undiagnosed malignancy or micro-metastatic disease, TB-500's **pro-migration and pro-angiogenic signaling** could *theoretically* support tumor progression. In people without cancer, available data do **not** show carcinogenicity, but the mechanistic overlap with metastatic biology justifies a **cautionary stance**, especially for those with a history of cancer, active malignancy, or elevated cancer risk.

### Potential Drug Interactions

- **ACE inhibitors:** May affect levels of the T $\beta$ 4 metabolite Ac-SDKP, which is normally degraded by ACE. Concurrent use could potentiate anti-fibrotic effects.
- **Anticoagulants and antiplatelet agents:** Theoretical interaction due to T $\beta$ 4 release from platelets and potential effects on platelet function. Monitor closely.
- **Immunosuppressive agents:** T $\beta$ 4 has immunomodulatory properties that may interact with immunosuppressive therapy. Clinical significance unknown.
- **Other angiogenic agents:** Potential additive or synergistic effects on blood vessel formation. Concurrent use warrants monitoring.

### Conclusions

We do not have human safety data for TB-500 from large, randomized trials. We infer short term safety from its widespread use. But short-term safety is not proof of long-term safety for continuous use. Long term concerns around angiogenesis and cancer promotion are legitimate.

Some researchers with low-to-moderate risk tolerance might choose to use TB-500 only for acute injury repair.

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